**GOVERNMENT OF KARNATAKA**
**KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD**
**WEIGHTAGE FRAMEWORK FOR MQP 3: II PU MATHEMATICS (35):2024-25**

| Chapter | CONTENT | Number of Teaching hours | PART A 1 mark | | PART B 2 mark | PART C 3 mark | PART D 5 mark | PART E | | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | MCQ | FB | | | | 6 mark | 4 mark | |
| 1 | RELATIONS AND FUNCTIONS | 9 | 1 | | | 1 | 1 | | | 9 |
| 2 | INVERSE TRIGONOMETRIC FUNCTIONS | 6 | 2 | | 2 | | | | | 6 |
| 3 | MATRICES | 9 | 1 | 1 | 1 | | 1 | | | 9 |
| 4 | DETERMINANTS | 12 | 1 | | 1 | | 1 | | 1 | 12 |
| 5 | CONTINUITY AND DIFFERENTIABILITY | 20 | 2 | 1 | 1 | 1 | 1 | | 1 | 17 |
| 6 | APPLICATION OF DERIVATIVES | 10 | 2 | 1 | 1 | 1 | | | | 8 |
| 7 | INTEGRALS | 22 | 1 | 1 | 1 | 1 | 1 | 1 | | 18 |
| 8 | APPLICATION OF INTEGRALS | 5 | | | | | 1 | | | 5 |
| 9 | DIFFERENTIAL EQUATIONS | 10 | 1 | | 1 | | 1 | | | 8 |
| 10 | VECTOR ALGEBRA | 11 | 2 | | | 2 | | | | 8 |
| 11 | THREE D GEOMETRY | 8 | 1 | | 1 | 1 | | | | 6 |
| 12 | LINEAR ROGRAMMING | 7 | | | | | | 1 | | 6 |
| 13 | PROBABILITY | 11 | 1 | 1 | | 2 | | | | 8 |
| | **TOTAL** | **140** | **15** | **5** | **9** | **9** | **7** | **2** | **2** | **120** |

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# Model Question Paper -3

## II P.U.C MATHEMATICS (35):2024-25

**Time : 3 hours** **Max. Marks : 80**

**Instructions :**

1) *The question paper has five parts namely A, B, C, D and E. Answer all the parts.*
2) *PART A has 15 MCQ's ,5 Fill in the blanks of 1 mark each.*
3) *Use the graph sheet for question on linear programming in PART E.*

**PART A**

**I. Answer ALL the Multiple Choice Questions** **15×1 = 15**

**1**. The element needed to be added to the relation R={(1,1), (1,3), (2,2),(3,3) } on A = {1, 2, 3} so that the relation is neither symmetric nor transitive

A) (2, 3) B) (3, 1) C) (1, 2) D) (3, 2)

**2**. The graph of the function $y = \cos^{-1} x$ is the mirror image of the graph of the function y = cosx along the line

A) x = 0 B) y = x C) y = 1 D) y = 0

**3**. The value of $\tan^{-1}(\sqrt{3}) + \sec^{-1}(-2)$ is equal to

A) π B) $\frac{2\pi}{3}$ C) $-\frac{\pi}{3}$ D) $\frac{\pi}{3}$

**4**. If A and B are matrices of order 3 × 2 and 2 × 2 respectively, then which of the following are defined

A) AB B) BA C) $A^2$ D) A + B

**5**. A square matrix $A$ is invertible if $A$ is

A) Null matrix B) Singular matrix
C) skew symmetric matrix of order 3 D) Non-Singular matrix

**6**. If $y = \sin^{-1}(x\sqrt{x})$, then $\frac{dy}{dx} =$

A) $\frac{1}{\sqrt{1-x^3}}$ B) $\frac{2\sqrt{x}}{3\sqrt{1-x^3}}$ C) $\frac{3\sqrt{x}}{2\sqrt{1-x^3}}$ D) $\frac{-3\sqrt{x}}{2\sqrt{1-x^3}}$.

**7**. If $y = x^a + a^x + a^a$ for some fixed $a > 0$ and $x > 0$, then $\frac{dy}{dx} =$

(A) $ax^{a-1} + a^x \log a + aa^{a-1}$ B) $ax^{a-1} + a^x \log a$

C) $ax^{a-1} + xa^{x-1} + aa^{a-1}$ D) $ax^{a-1} + a^x \log a + a^a$ .

**8**. Consider the following statements for the given function y=f(x) defined on an interval I and c∈ I, at x = c

I. $f'(c) = 0$ and $f''(c) < 0 \Rightarrow$ f attains local maxima
II. $f'(c) = 0$ and $f''(c) > 0 \Rightarrow$ f attains local minima
III. $f'(c) = 0$ and $f''(c) = 0 \Rightarrow$ f attains both maxima and minima

A) I and II are true B) I and III are true

C) II and III are true D) all are false

**9**. If each side of a cube is x units, then the rate of change of its surface area with respect to side is

A) 12x B) 6x C) $6x^2$ D) $3x^2$

**10.** Statement 1: The anti-derivative of $\left(\frac{1}{\sqrt{1+x^2}}\right)$ with respect to x is

$$\frac{x}{2}\sqrt{1+x^2}+\frac{1}{2}\log\left|x+\sqrt{1+x^2}\right|+C.$$

Statement 2**:** The derivative of $\frac{x}{2}\sqrt{1+x^2}+\frac{1}{2}\log\left|x+\sqrt{1+x^2}\right|+C$ with respect to x is $\frac{1}{\sqrt{1+x^2}}$.

A) Statement 1 is true, and Statement 2 is false.
B) Statement 1 is true, and Statement 2 is true, Statement 2 is correct explanation for Statement 1
C) Statement 1 is true, and Statement 2 is true, Statement 2 is not a correct explanation for Statement 1
D) Statement 1 is false, and Statement 2 is true.

**11**. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^2+\sin\left(\frac{dy}{dx}\right)+1=0$ is

A) 2 B) 3 C) 5 D) not defined

**12**. The position vector of a point which divides the join of points with position vectors $3\vec{a}-2\vec{b}$ and $\vec{a}+\vec{b}$ externally in the ratio 2 : 1 is

A) $\frac{5\vec{a}}{3}$ B) $4\vec{a}-\vec{b}$ C) $4\vec{b}-\vec{a}$ D) $2\vec{a}+\vec{b}$

**13.** If a vector $\vec{a}$ makes angles with $\frac{\pi}{3}$ with $\hat{\imath}$ *and* $\frac{\pi}{4}$ with $\hat{\jmath}$ and an acute angle $\theta$ with $\hat{k}$, then θ is

A) $\frac{\pi}{6}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{2}$

**14**. Find the angle between the lines whose direction ratios are $a, b, c$ and $b-c, c-a, a-b$ is

A) $45^0$ B) $30^0$ C) $60^0$ D) $90^0$

**15**. If A and B are two independent events such that $P(A) = \frac{1}{4}$ and $P(B) = \frac{1}{2}$ then P(neither A nor B)

A) $\frac{1}{3}$ B) $\frac{3}{8}$ C) $\frac{7}{8}$ D) $\frac{1}{2}$ .

**II. Fill in the blanks by choosing the appropriate answer from those given in the bracket (-2, $\frac{5}{2}$, 0, 1, 2, $\frac{3}{2}$ ) 5×1 = 5**

**16**. The number of all possible orders of matrices with 13 elements is ____

**17**. If $y$ = 5 cos $x$ – 3 sin $x$, then $\frac{d^2y}{dx^2} + y =$_____

**18**. If the function $f$ given by $f(x) = x^2 + ax + 1$ is increasing on [1, 2], then the value of 'a' is greater than _____

**19**. $\int_1^2 |x|\,dx =$_____

**20**. If A and B are any two events such that P(A) + P(B) – P(A and B) =P(A), then P(A | B) is_____

**PART B**

**Answer any SIX questions** **6 × 2=12**

**21**. Write the simplest form of $\tan^{-1}\left(\sqrt{\frac{1-cosx}{1+cosx}}\right)$, $0 < x < \pi$.

**22.** Prove that $2\sin^{-1}\frac{3}{5} = \tan^{-1}\frac{24}{7}$.

**23**. If F(x)= $\begin{bmatrix} cosx & -sinx & 0 \\ sinx & cosx & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then show that F($x$) F($y$) = F($x + y$).

**24**. Find the equation of line joining (1, 2) and (3, 6) using determinants.

**25**. Differentiate $x^{\sin x}$, $x > 0$ with respect to x.

**26**. Find the intervals in which the function $f$ given by $f(x) = x^2e^{-x}$ is increasing.

**27.** Find $\int (x^2+1)\log x\,dx$ .

**28.**Verify the function $y$ =mx is the solution of $\frac{dy}{dx} - y = 0$, x ≠ 0.

**29.** Find the distance between the lines $\vec{r} = \hat{\imath} + 2\,\hat{\jmath} - 4\,\hat{k} + \lambda\,(2\hat{\imath} + 3\,\hat{\jmath} + 6\,\hat{k})$ and $\vec{r} = 3\hat{\imath} + 3\hat{\jmath} - 5\,\hat{k} + \mu(2\hat{\imath} + 3\,\hat{\jmath} + 6\,\hat{k})$ .

## PART C

**Answer any SIX questions** **6 × 3= 18**

**30**. Let $f$: X →Y be a function. Define a relation R in X given by R = {$(a, b)$: $f(a) = f(b)$}. Examine whether R is an equivalence relation or not.

**31**. If $x^3 + x^2y + xy^2 + y^3 = 81$, then find $\frac{dy}{dx}$.

**32**. The length $x$ of a rectangle is decreasing at the rate of 3 cm/min and the width $y$ is increasing at the rate of 2 cm/min. When $x = 10$ cm and $y = 6$ cm, find the rate of change of the perimeter of the rectangle.

**33**. Find the integral of $\frac{1}{a^2 + x^2}$ with respect to x .

**34**. If the vertices $A$, $B$ and $C$ of a triangle are $(1, 2, 3)$, $(-1, 0, 0)$ and $(0, 1, 2)$ respectively, then find the angle $\angle ABC$.

**35.** Find the area of the rectangle, whose vertices are $A\left(-\hat{i}+\frac{1}{2}\hat{j}+4k\right)$, $B\left(\hat{i}+\frac{1}{2}j+4k\right)$, $C\left(\hat{i}-\frac{1}{2}j+4k\right)$ and $D\left(-\hat{i}-\frac{1}{2}j+4k\right)$.

**36.** Find the vector equation of the line passing through the point (1, 2, – 4) and perpendicular to the two lines: $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.

**37**. An urn contains 5 red and 5 black balls. A ball is drawn at random, its colour is noted and is returned to the urn. Moreover, 2 additional balls of the colour drawn are put in the urn and then a ball is drawn at random. What is the probability that the second ball is red.

**38**. Three coins are tossed simultaneously. Consider the Event E 'three heads or three tails', F 'at least two heads and G 'at most two heads'. Of the pairs (E,F), (E, G) and (F, G), which are independent? Which are dependent?

## PART D

**Answer any FOUR questions** **4 × 5 = 20**

**39**. Let $f$: N →Y be a function defined as $f(x) = 4x + 3$, where, Y = {$y \in$ N: $y = 4x + 3$ for some $x \in$ N}. Show that $f$ is invertible. Find the inverse of f.

**40**. If $A = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$, calculate AB, AC and A(B + C). Verify that $A(B+C) = AB + AC$.

**41**. Solve the following system of linear equations by matrix method:

$2x + y + z = 1$, $x - 2y - z = \frac{3}{2}$ and $3y - 5z = 9$.

**42**. If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, find $\frac{d^2y}{dx^2}$.

**43.** Find $\int \frac{x^4}{(x-1)(x^2+1)} dx$.

**44**. Find the area of the region bounded by the line $y = 3x + 2$, the $x$-axis and the ordinates $x = -1$ and $x = 1$ by integration method.

**45.** Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point $(x, y)$ is equal to the sum of the ordinates of the point.

**PART E**

**Answer the following questions:**

**46**. (a) Prove that $\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\int_{0}^{a} f(x)\,dx, & \text{if f(x) is even} \\ 0, & \text{if f(x) is odd} \end{cases}$

and evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \, dx$

**OR**

Solve the following linear programming problem graphically:

Minimize and maximize $Z = x + 2y$, subject to constraints

$x + 2y \geq 100,\ 2x - y \leq 0,\ 2x + y \leq 200\ and\ x, y \geq 0.$

**6**

**47.** If matrix A= $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ satisfying $A^3-6A^2+9A-4I=O$, then evaluate $A^{-1}$.

**OR**

If $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & If\ x \neq \frac{\pi}{2} \\ 3, & if\ x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, find k . **4**

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